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COLLEGE OF SCIENCE AND TECHNOLOGY

SCHOOL OF ICT

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MODULE: MOBILE APPLICATION SYSTEM AND DESIGN

TOPIC: HARD-CODED MULTI-SCREEN TRAVEL APP UI LAB 3 REPORT

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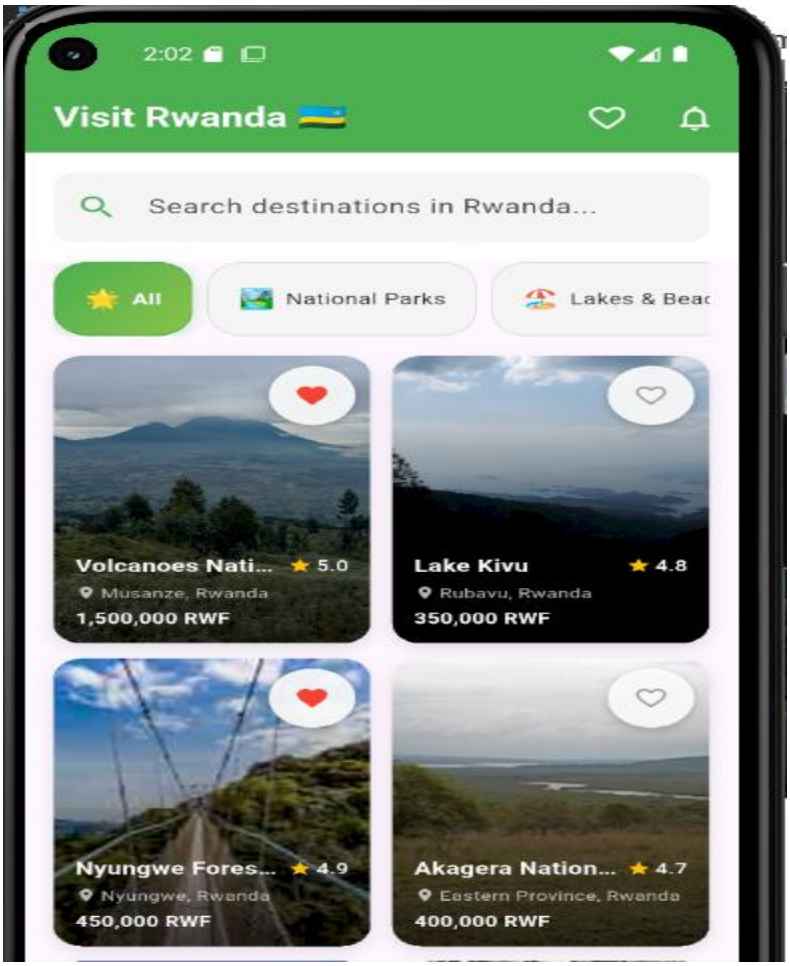
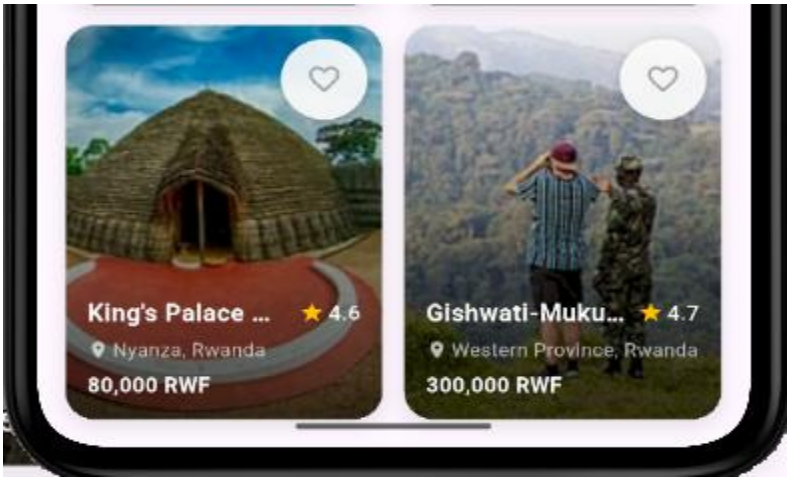
SCREENSHOTS OF ALL SCREENS AND EXPLANATIONS OF LAYOUT CHOICES

1.Home Screen





Home Screen Layout Explanation

The Home Screen functions as the main dashboard of the Visit Rwanda application. It was designed following Material Design principles with a focus on visual hierarchy, scannability, and ease of navigation.

App Bar: **Positioned at the top with the app title "Visit Rwanda" and action icons for favorites and notifications. This follows standard mobile app conventions, making the interface immediately familiar to users. The Rwandan flag emoji reinforces the app's cultural theme.**

Search Bar: **Placed directly below the App Bar, the search field features a light grey background and search icon, creating visual separation from the content below. While non-functional per lab requirements, its placement follows the expected F-shaped reading pattern where users look for search at the top of the screen.**

Category Chips: **Horizontal scrolling categories allow users to filter destinations by type (National Parks, Lakes, etc.). This layout choice saves vertical screen space while providing quick access to filtering. The "All" chip is always visible to reset filters, and selected chips change color to provide visual feedback.**

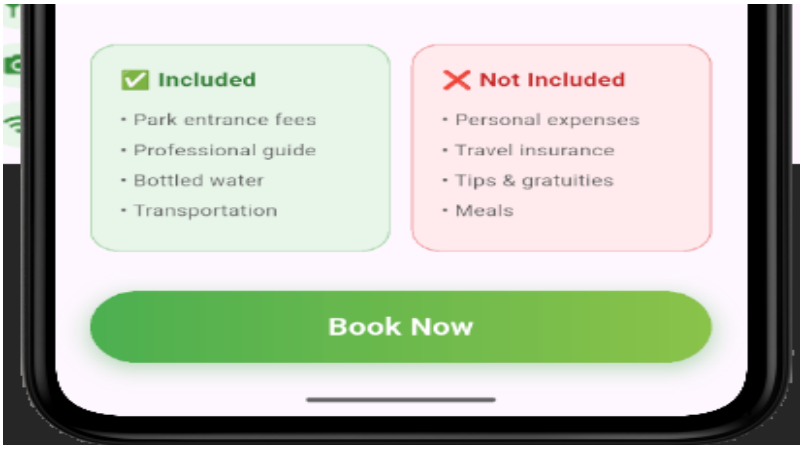
Destination Grid: **A 2-column grid displays destination cards efficiently. This configuration was chosen because:**

- **3 columns would make images too small on phone screens**
- **1 column would require excessive scrolling**
- **12px spacing between cards creates visual rhythm**

Destination Cards: **Each card uses a Stack widget to layer multiple elements:**

- **Background image creates visual appeal**
- **Gradient overlay ensures text readability**
- **Favorite button positioned at top-right for easy access**
- **Text content at bottom shows name, location, rating, and price in RWF**

2. Detail Screen



Detail Screen Layout Explanation

The Detail Screen provides comprehensive information about a selected destination. It is designed as a scrollable page that presents all relevant details in a logical, easy-to-digest format.

Header Image: **A full-width hero image (300px height) creates immediate visual impact and emotional connection with the destination. A gradient overlay from transparent to black ensures that text overlaid on the image remains readable regardless of the image's colors. Back and favorite buttons are positioned in the top corners following platform conventions, making navigation intuitive.**

Rating and Price Section: **This information is displayed in a side-by-side layout to save vertical space. The rating appears in a green pill-shaped container with a star icon, drawing attention to the quality score. The price is displayed in large bold text on the right with "per person" clarification below, creating clear visual hierarchy and ensuring users understand the cost structure.**

Description Section: **A clear section header helps users quickly identify this content. The body text uses increased line height (1.5) for improved readability of longer passages, and grey text color reduces visual weight compared to more important elements like prices and ratings.**

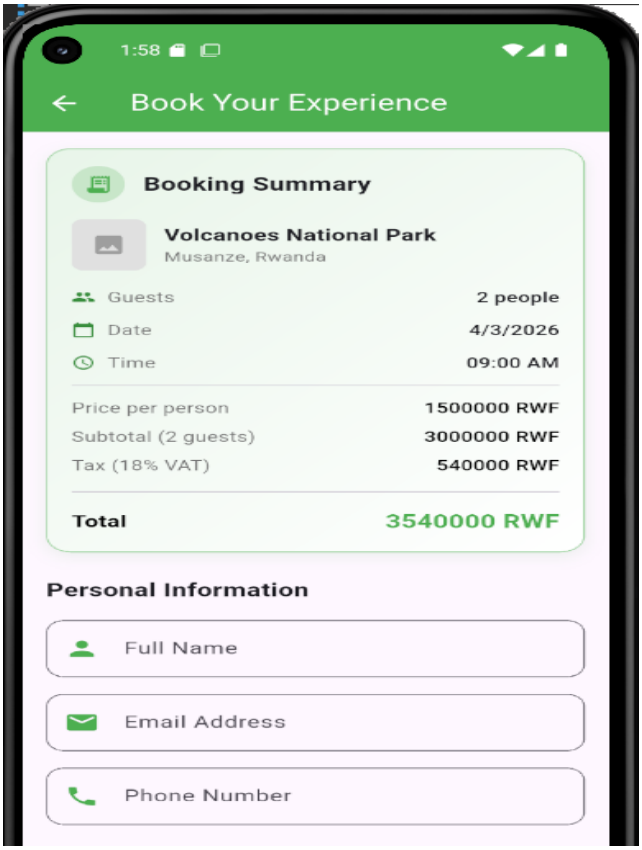
Highlights Section: **Key features are presented as icon-text pairs with circular green backgrounds. This format is more scannable than text alone, and the consistent use of green reinforces brand identity throughout the app. Each highlight appears in a vertical list with consistent spacing for a polished, professional look.**

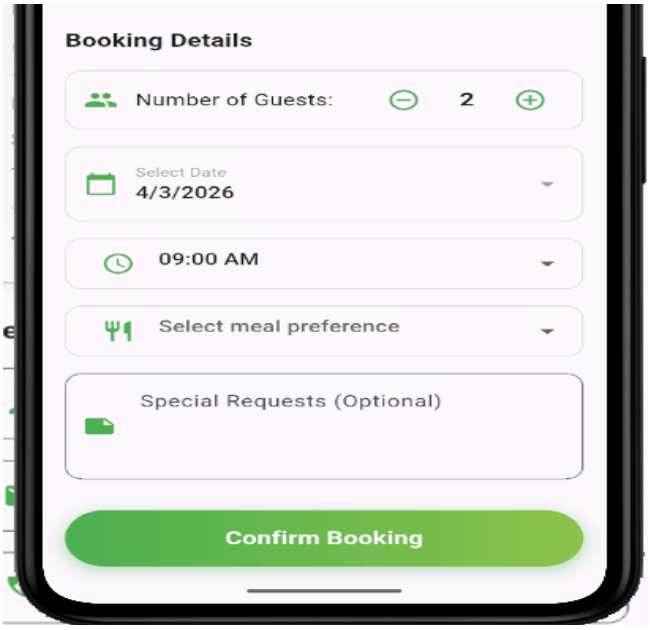
Included/Not Included Section: **A 50/50 split layout using two Expanded widgets allows users to quickly compare what's included versus what's not. Color coding (green for included, red for not included) provides intuitive visual cues. Bullet points make individual items easy to scan, and colored backgrounds with matching borders create cohesive visual units.**

Book Now Button: **The primary call-to-action appears as a full-width gradient button with rounded corners and a subtle shadow. Its prominent position at the bottom of the content follows the natural reading flow (information first, action second). The button meets touch target guidelines (55px height) and uses high-contrast white text for maximum visibility.**

Scrollable Structure: **The content after the header image is wrapped in a SingleChildScrollView, ensuring all information fits on any screen size while keeping the hero image fixed. Consistent 20px padding throughout creates breathing room around content and maintains visual rhythm.**

3. Booking Screen





Booking Screen Layout Explanation

The Booking Screen finalizes the reservation process by collecting user information and displaying a complete cost summary. It follows a form-style layout with clear sections and validation.

Booking Summary Card: **Positioned at the top of the screen, this prominent card displays all booking details including destination, dates, guest count, and a transparent price breakdown. The light green gradient background and subtle shadow make it stand out while maintaining the app's color theme. Showing the total cost upfront before collecting user information ensures price transparency and builds trust.**

Form Organization: **The screen is divided into two logical sections: "Personal Information" and "Booking Details." Each section has a bold header, and consistent 24px spacing between sections helps users mentally separate different types of information. This grouping reduces cognitive load and makes long forms feel more manageable.**

Form Fields: **Each TextFormField uses an outlined border with 12px rounded corners, matching the app's modern aesthetic. Prefix icons provide visual cues about expected input (person icon for name, email icon for email, etc.). Floating labels save space while maintaining context, and the green focus state provides clear feedback when a field is active.**

Validation: **Every required field includes a validator function that displays error messages below the field when validation fails. The Form widget with a GlobalKey allows all fields to be validated at once when the Confirm button is pressed, ensuring complete data entry before proceeding.**

Guest Counter: **A custom counter with +/- buttons provides an intuitive way to adjust guest count. The minus button is disabled when at minimum (1 guest), preventing invalid selections. The current count is displayed prominently in a 40px wide container between the buttons.**

Date and Time Selection: **The date picker uses a tappable container that looks like a dropdown, showing the currently selected date with a calendar icon. When tapped, it opens the native Android/iOS date picker for a familiar user experience. Time selection uses a DropdownButtonFormField with predefined time slots.**

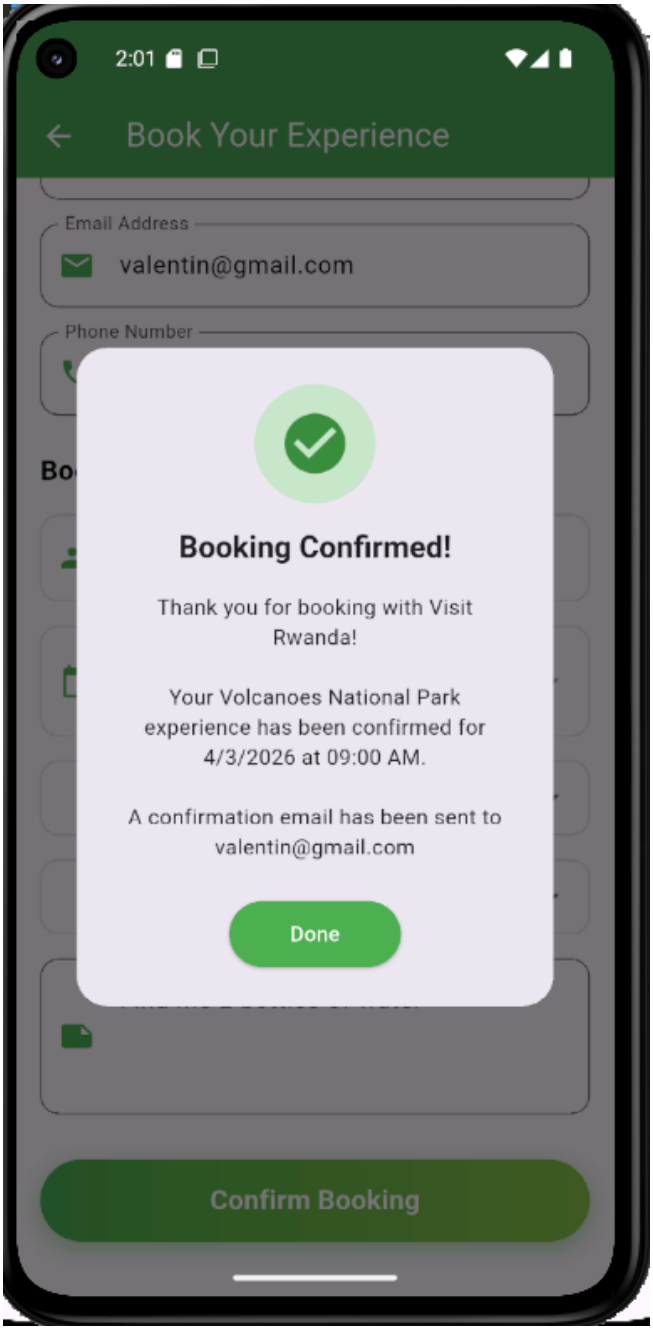
Success Feedback: **After successful validation, an AlertDialog appears with a large checkmark icon in a green circular background. The dialog confirms the booking details and provides a single "Done" button that returns users to the home screen, completing the booking flow.**

Button Consistency: **The Confirm Booking button uses the same gradient styling as the Detail Screen's Book Now button, creating visual consistency throughout the app. Its full-**

width design and prominent position at the bottom of the form naturally follow the completion of data entry.

This layout successfully meets the lab requirement for a booking screen by combining a clear cost summary with an organized, validated form structure.

4.Success dialog



Success Dialog Layout Explanation

Success Dialog: After form validation, an AlertDialog appears with a green checkmark icon, "Booking Confirmed!" title, and a summary of the booking details including destination, date, time, and confirmation email. A single "Done" button returns users to the home screen, completing the booking flow with clear positive feedback.

COMPLETE LIST OF WIDGETS USED IN TRAVEL APP

Numbers	Widget Name	Used In	Purpose
1	**MaterialApp**	main.dart	Root app widget with theming
2	**Scaffold**	All screens	Basic material design layout structure
3	**AppBar**	All screens	Top bar with title and actions
4	**Container**	Multiple files	Box with decoration, padding, margins
5	**Column**	Multiple files	Arranges children vertically
6	**Row**	Multiple files	Arranges children horizontally
7	**Text**	Multiple files	Displays text content
8	**Icon**	Multiple files	Displays icons
9	**IconButton**	Multiple files	Clickable icons in AppBar and cards

Numbers	Widget Name	Used In	Purpose
10	**ElevatedButton**	Detail/Booking screens	Styled buttons (Book Now, Confirm)
11	**TextField**	Home/Booking screens	Text input fields
12	**TextFormField**	Booking screen	Form field with validation
13	**Form**	Booking screen	Groups form fields for validation
14	**GestureDetector**	Destination card	Detects taps on cards
15	**Stack**	Destination card	Layers widgets (image, gradient, text)
16	**Positioned**	Destination card	Positions widgets in Stack
17	**Positioned.fill**	Destination card	Fills entire Stack area
18	**ClipRRect**	Destination card	Rounds corners of images
19	**Image.asset**	Destination card	Displays images from assets
20	**DecoratedBox**	Destination card	Applies gradient overlay
21	**LinearGradient**	Multiple files	Gradient backgrounds
22	**BoxDecoration**	Multiple files	Decorates containers with colors, borders

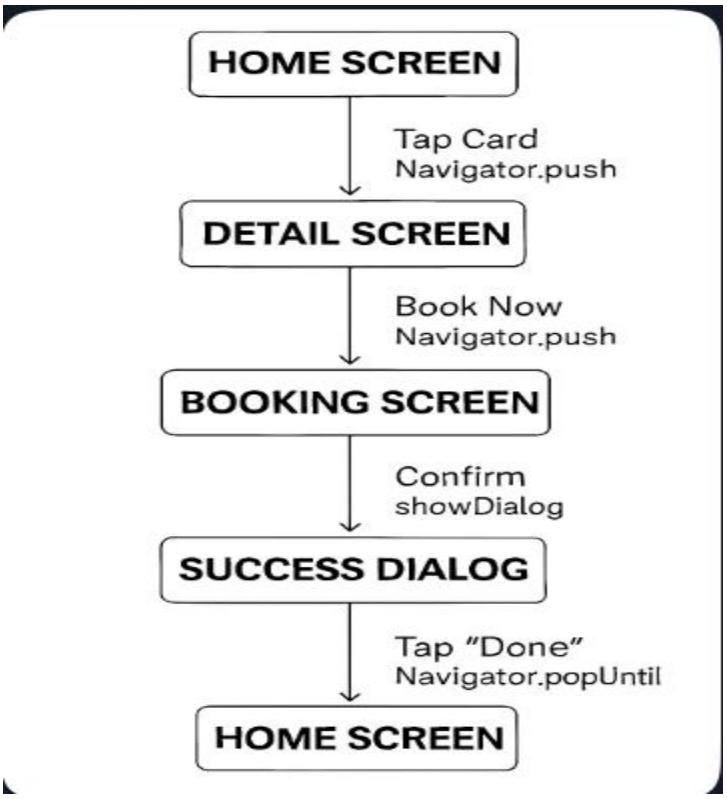
Numbers	Widget Name	Used In	Purpose
23	**BoxShadow**	Multiple files	Adds shadow effects
24	**BorderRadius**	Multiple files	Rounds corners
25	**EdgeInsets**	Multiple files	Adds padding and margins
26	**SizedBox**	Multiple files	Adds fixed spacing
27	**Expanded**	Home/Booking screens	Takes remaining space in Row/Column
28	**ListView**	Home screen	Horizontal scrolling categories
29	**ListView.builder**	Home screen	Builds list efficiently
30	**GridView.builder**	Home screen	Grid of destination cards
31	**SliverGridDelegateWithFixedCrossAxisCount**	Home screen	Defines grid layout
32	**Center**	Multiple files	Centers child widgets
33	**SingleChildScrollView**	Detail/Booking screens	Makes content scrollable
34	**Padding**	Multiple files	Adds padding around widgets
35	**Divider**	Booking summary	Horizontal line separator

Numbers	Widget Name	Used In	Purpose
36	**DropdownButtonFormField**	Booking screen	Dropdown for time/meal selection
37	**DropdownMenuItem**	Booking screen	Items in dropdown
38	**AlertDialog**	Booking screen	Success dialog after booking
39	**SnackBar**	Multiple files	Temporary messages
40	**CircularProgressIndicator**	Home screen	Loading indicator
41	**InkWell**	Booking screen	Date picker trigger
42	**FutureBuilder**	*(if used)*	Async data handling
43	**MediaQuery**	*(for responsiveness)*	Gets screen dimensions

CUSTOM REUSABLE WIDGETS CREATED

Numbers	Custom Widget	File	Purpose
44	**DestinationCard**	destination_card.dart	Reusable destination card with image, gradient, text
45	**CategoryChip**	category_chip.dart	Reusable category chip with icon and selection state
46	**BookingSummaryCard**	booking_summary_card.dart	Reusable booking summary with price breakdown

NAVIGATION FLOW DIAGRAM



The diagram shows how a user moves through our app:

1. User starts at **Home Screen** and taps a destination card → goes to **Detail Screen**
2. On Detail Screen, user taps "Book Now" → goes to **Booking Screen**
3. On Booking Screen, user fills form and taps "Confirm" → a **Success Dialog** pops up
4. User taps "Done" on the dialog → goes back to **Home Screen**

CONCLUSION

In this lab, our group successfully developed a **multi-screen travel app UI** using Flutter, featuring hard-coded data for Rwandan destinations. The application includes three main screens: a **Home Screen** with a complex dashboard layout, a **Detail Screen** with scrollable content and booking navigation, and a **Booking Screen** with form validation and success feedback.